

Hamza Mohammed Hassanain

Egypt | +20 12 7247 3302 | hamzahassanain067@gmail.com | linkedin.com/in/hamza-hassanain-ha067 | github.com/HamzaHassanain
hamzahassanain.github.io

SUMMARY

Software Engineer at **Repovive (YC F'26)**, building the contest judge and core infrastructure for **10,000+ users**. **AWS Certified Solutions Architect**, **3× ICPC ACPC finalist**, and author of ICPC JCPC 2025 problem *G. Queries On Strings*. I contribute to **GitLab** and **LLVM** and write deep dives on dev.to, like *What Every Programmer Should Know About Memory*.

EDUCATION

B.Sc. in Computer and Information Sciences — Suez Canal University 2022 – 2026
Graduation Project: [SigmaLoop](#), an AI tutor & curriculum engine (Grade: A+) Ismailia, Egypt
Coursework: Data Structures & Algorithms, Operating Systems, Networks, Databases, Distributed Systems, Cloud Computing

EXPERIENCE

Repovive — YC F'26 Nov 2025 – Present
Software Engineer (Full Time) United States (Remote)

Repovive is a **Y Combinator-backed** science expert community where experts solve unsolved scientific problems, starting with theoretical computer science and math proof reasoning. We run **live programming and math contests**.

- Joined as the **first engineer** and built the core systems from the early days through **YC** and **10,000+ users**.
- Onboarded and supported new engineers, senior hires included, guiding them through the codebase and projects.
- Led design of the distributed judge (**judge1** on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** test cases.
- Collaborated with the team to build AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build).
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped fixes.

OPEN SOURCE

GitLab Nov 2025 – Jan 2026
Open Source Contributor [MRs](#)

- Added typed group integrations to the Go SDK: Slack, Jira, MS Teams, Mattermost, Harbor, +3 [!2692](#), [!2675](#), [!2670](#)
- Added Harbor and Mattermost integration resources to the Terraform provider, plus a group SAML attribute [!2881](#), [!2909](#)
- Hardened **Json.safe_parse** and fixed Vue lint issues in the main GitLab codebase [!221163](#), [!221158](#)

LLVM Nov 2025 – Feb 2026
Compiler Contributor [PRs](#)

- Added constexpr evaluation of the X86 **cvtpd2ps** intrinsics in Clang [#169980](#)
- Lowered AArch64 and ARM CLS to **ISD::CTLs**, replacing a manual expansion [#178885](#), [#178430](#)
- Added LLDB SWIG Python extension tests for **SBTarget** [#173473](#)

ACHIEVEMENTS & CERTIFICATIONS

- Certifications:** AWS Certified Solutions Architect – Associate (SAA-C03) credly.com/verify
- Competitive Programming:** ICPC ACPC 3× Finalist (2024–2026); 17th of ~500 teams at ECPC 2024 icpc.global
- Problem Author:** *G. Queries On Strings* (JCPC 2025), *F. Graph Tiling* (Repovive Round 3) codeforces.com | repovive.com

PROJECTS

Polyman CLI | *Creator & Maintainer* [npm](#) | [GitHub](#)
Polyman is an open-source CLI that lets problem setters and **AI agents** build and test problems locally, then push to Polygon.

- One **verify** checks everything before upload: tests, validator, checker, and each solution's expected verdict.
- Built an **agentic workflow**: Claude Code/Codex builds the problem, loops on **verify --json** until green, then pushes.
- Grown from user feedback and outside PRs; **1,600+ npm downloads**, launch post **+332** on Codeforces.

CP Judge on AWS | *Reference Architecture* [GitHub](#)

- Designed Judge0 **isolate** on ECS-on-EC2 with a Fargate orchestrator and a Spot ASG scaled on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

SKILLS

Languages: C++, Go, Rust, Python, TypeScript, SQL | Node.js, Express.js, Next.js, React | MongoDB, PostgreSQL
Cloud & Concepts: AWS, Docker, Nginx, Redis, Terraform, CI/CD | System Design, Distributed Systems, Message Queues